

07 . 1

What is meant by the term phenotype?

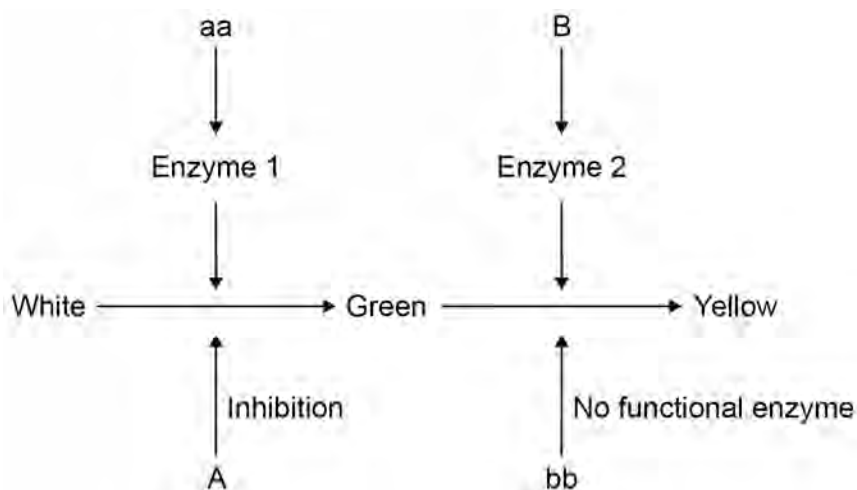
[2 marks]

07 . 2

The inheritance of fruit colour in summer squash plants is controlled by two genes, **A** and **B**. Each gene has two alleles.

Figure 7 shows the interaction of these two genes in controlling fruit colour in summer squash plants.

Figure 7



Name the type of gene interaction shown in **Figure 7**.

[1 mark]

07 . 3

What fruit colour would you expect the following genotypes to have?

[1 mark]

AAbb _____

aaBB _____

07 . 4

Genes **A** and **B** are **not** linked.

Complete the genetic diagram to show all the possible genotypes and the ratio of phenotypes expected in the offspring of this cross.

[3 marks]

Genotypes of parents **aabb** × **AaBb**

Genotypes of offspring _____

Phenotypes of offspring _____

Ratio of phenotypes _____

07 . 5

A population of summer squash plants produced only green and yellow fruit. The percentage of plants producing yellow fruit in this population was 36%.

Use the Hardy–Weinberg equation to calculate the percentage of plants that were heterozygous for gene **B**.

[2 marks]

Answer = _____ %

Turn over for the next question

Turn over ►

